



OPERATIONAL MANUAL

INSTRUCTION MANUAL
AIR SHOWER CONTROLLER
MODEL NO: AS153
VERSION: 1.0



NEURAL INDUSTRIAL AUTOMATION PVT LTD.,
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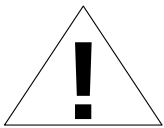
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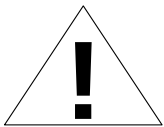
Email: [sales@ neuralindustrialautomation.in](mailto:sales@neuralindustrialautomation.in)

Dear Customer,

Thank you for choosing a NEURAL instrument, please read this operating manual carefully before you start using the instrument. This manual will provide you with all the necessary information for the correct use of the instrument as well as a precise idea of its versatility in a wide range of applications.

**Warning:**

Warnings must be followed carefully to avoid serious bodily injury.

**CAUTION:**

Cautions must be observed to avoid minor injury to yourself or damage to your equipment.

CAUTION

One three way connector is provided for power connection i.e. Line, Neutral & Earth. (100-240V AC 50/60Hz.)



Do not Switch ON power to the instrument before plugging connectors in to the terminals.



Carefully connect the connectors in the specified terminals if not it may cause damage for the instrument.

Instrument SI. No :

STANDARD PACK

1) Instruction Manual	---	1 No	☐
2) AS153 Instrument	---	1 No	☐

Checked By :

Date :

1. Description:

The air shower is a supporting device for the clean workshops and the assembly clean room, is used to blow the dusts attached to the surfaces of the people and objects entering the clean workshop, and also functions as an air brake to prevent non-purified air from entering the clean area. It is an effective device to clean up the surfaces of the persons and the objects and to prevent the outdoor air from invading the clean area. It has strong versatility, can be used along with all clean rooms and clean workshops, and is an ideal choice to establish a dust-free workshop.

The Air Shower interlocking system controls the sequential locking and release of the doors & control the Blower. The two doors are interlocked such that two doors cannot be open at the same time, to prevent pressure losses and cross contamination.

Our Air Shower interlocking system achieve this by preventing simultaneous access of door by ensuring that when one door for a Air Shower interlocking system is open then other door is closed before opening of door.

1. When Door 1 is opened the display on the Door 2 will indicate “WAIT” & person access the door by pressing push button.
2. Both Doors are provided with EM Locks

2. Controller Features:

- Advanced Micro controller based.
- Supports Maximum of 2 doors
- Supports Electro magnetic locks/Electric strikers
- Door status indication with high intensity Bar LEDs
- Status indication: GO, WAIT on push button station.
- Digital Display with soft membrane keypad.
- Status indication: OPEN (Door 2 open), Repeat, Mode of operation: Manual, Auto on digital display module.

- User programmable Blower ON time variable from 0 – 255 Sec.
- Buzzer output for door 2 is open.
- CFL lamp will be ON automatically when person access the door

Specifications:

GENERAL

- Electro Magnetic Lock Source : 24V DC/ 250 mA

Door Control by Relay:

RELAYS:

- Contact material : Silver Alloy.
- Contact Rating : 230V AC / 2 A.
- No. of Operations : Mechanical : 10^7 , Electrical : 10^5 .

ENCLOSURE:

Air Shower Controller

- Enclosure type : Power coating Box
- Mounting : Wall

Display Module

- Enclosure type : Northwest SS Bezel
- Mounting : Panel

POWER SUPPLY:

- Power : 100 - 280V, AC 50 Hz
- Wattage : 80W

ENVIRONMENTAL:

- Operating Temperature : 0 to 50°C.

3. Push Button Station

Features:

- ✓ Elegant look North-west SS Face plate.
- ✓ Indication facility for Door entry permitted / prohibited.
- ✓ Support an entry switch for opening the door.
- ✓ Long Life BAR LED for door open/close indication.
- ✓ Elegant look North-west SS Face plate with appropriate curvature to prevent resting of dust & powder.
- ✓ Easy mounting arrangement for fixing on Doorframes or on Wall.



Specifications:

GENERAL:

- Keypad : Soft touch Tactile Membrane Key
- Status LED's : WAIT Message for Door Open / Entry Prohibited.
GO Message for Door Close / Entry Permitted.
- Digital Input (LED) : 24V DC
- Life : 100, 000 Cycles Min.

ENCLOSURE:

- Enclosure type : Legrand Silver painted plastic Bezel/Face plate.
- Mounting : Flush mounting.
- Dimension : 3"X3" Single Module.

POWER:

Supply Voltage : 24V/50mA DC

4. Electromagnetic locks:

A **magnetic lock** is a simple locking device that consists of an electromagnet and armature plate. By attaching the electromagnet to the door frame and the armature plate to the door, a current passing through the electromagnet attracts the armature plate holding the door shut. The magnetic lock relies upon some of the basic concepts of electromagnetism. Essentially it consists of an electromagnet attracting a conductor with a force sufficiently large enough to prevent the door from being opened.



5. OPERATING MODE: Normally Close

Jumper setting:

Normally Close: Jumper position: J3 is Open

MANUAL MODE:

In this mode, initially both doors are in locked condition. Door access is by using push button. Status LED's provided on the push button station indicate whether a person can access the door or not.

GO = GO LED glow indicates that a person can access the door.

WAIT = WAIT LED glow indicates that a person cannot access the door.

Whenever a person enters the Air shower through door1 by pressing the push button, the shower will be ON when door1 is closed & both doors will be locked. The display shows the decrementing of shower on time. In this mode, repeat of shower can be possible by pressing the repeat key when the display is blinking. After completion of blower on time, door2 will be released up to 10 sec. whenever the emergency button is pressed, both doors will be unlocked up to 8 sec and GO, WAIT indication will be toggled.

AUTO MODE:

In this mode, initially both doors in locked condition. Door access by using push button. Status LED's provided on the push button station indicates whether person can access the door or not.

GO = GO LED glow indicates that person can access the door.

WAIT = WAIT LED glow indicates that person cannot access the door.

Whenever person enter in to the Air shower through door1 by pressing push button, Shower will be ON when door1 is closed & both doors will be locked. Display shows decrementing of shower on time. In this mode repeat of shower is not possible. After completion of blower on time door2 will be released up to 10 sec. whenever emergency button pressed both door will be unlocked up to 8 sec and GO, WAIT indication will be toggled.

6. Status Indication on Digital Display Panel :

MAN = Indicate that operating mode is manual.

AUTO = Indicate that operating mode is Auto.

EXIT OPEN = Indicate Door 2 (Exit) is opened

Emerg. = Indicates emergency condition

Repeat = Indicates repeating of shower in manual mode

Seven seg. Display = Indicates that Blower on time.

**7. Programming Mode:****a. Blower ON time setting:**

Sno	Description	Key to be pressed	Display	Action	Note
1.	For Setting the Blower ON time	SET	SET	Press ▲ key or ▼ key to set time from 0 – 255 sec	
2.	Accepting the Blower ON time	OK	025		

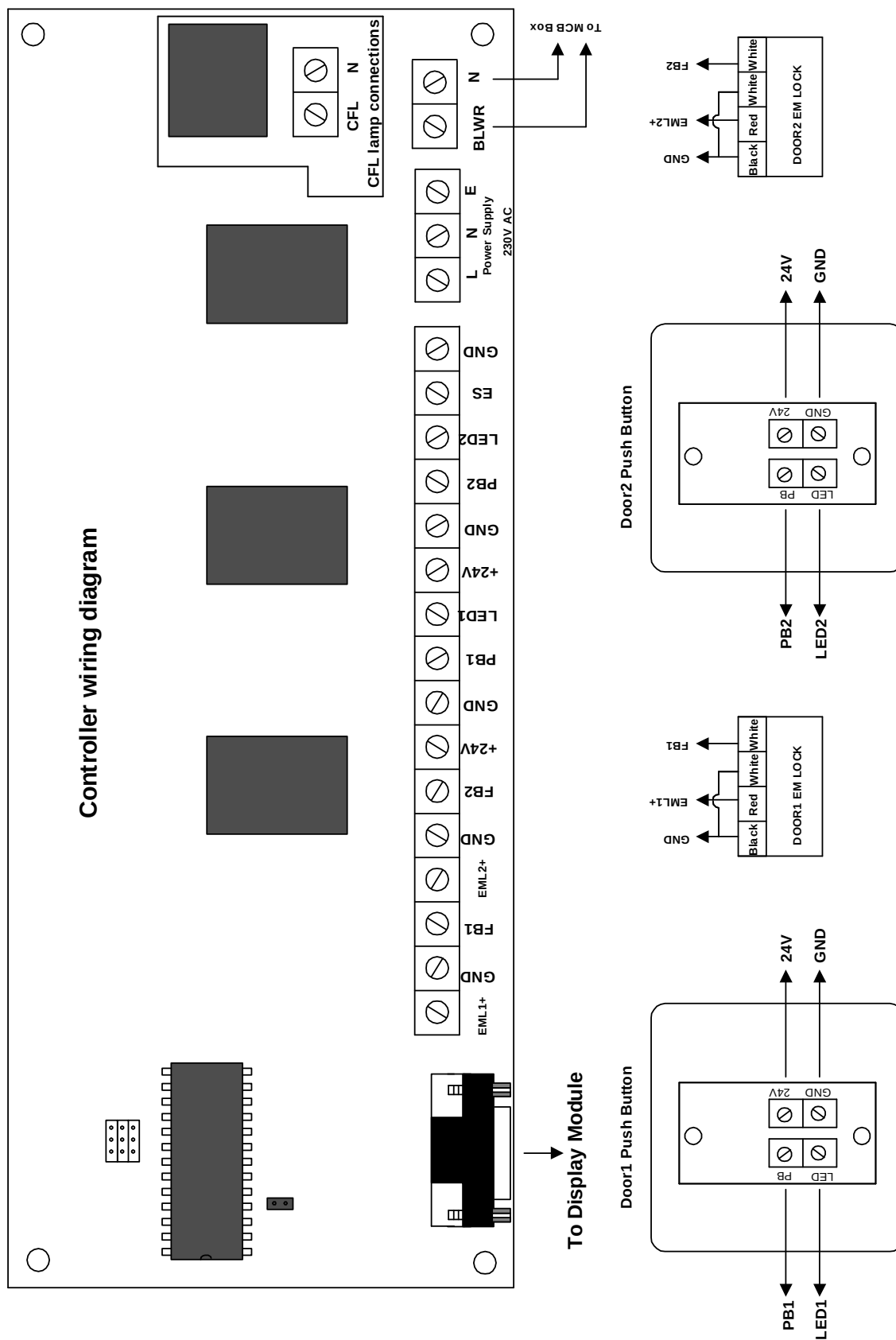
b. Operating Mode setting:

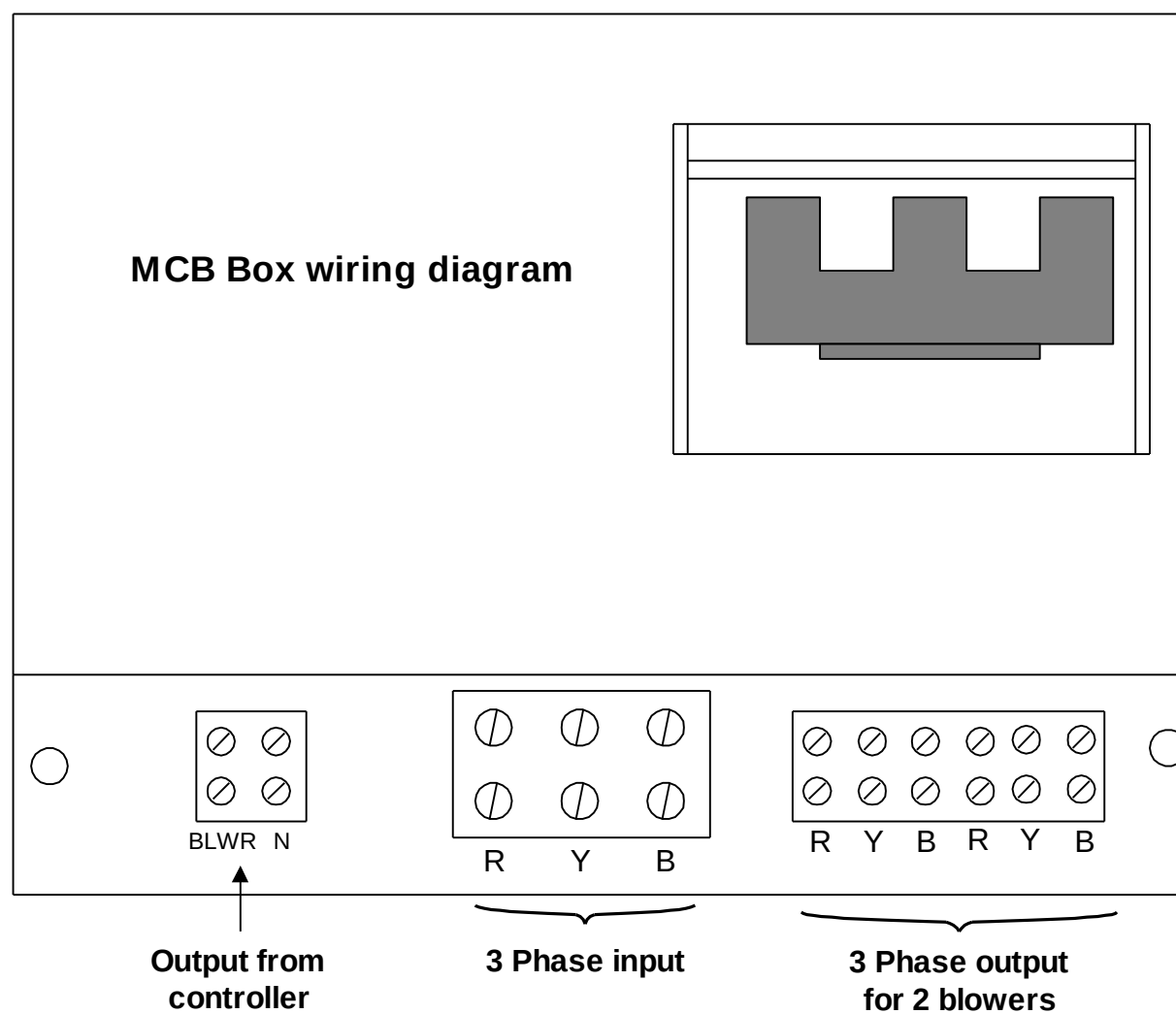
Sno	Description	Key to be pressed	Display	Action	Note
1.	For Setting the Operating Mode	SET	Set	Press  to set operating mode	
2.	Accepting the Operating Mode	OK	 (OR) 		

c. Emergency Mode:

When press Emergency button, BAR LED's on Push button will be toggle indicates that emergency situation, in this situation both doors in un locked condition up 20 sec.

8. Wiring diagram:





9. MECHANICAL DRAWINGS:

Fig (9.1) Figure for Installation of Air Shower Controller:

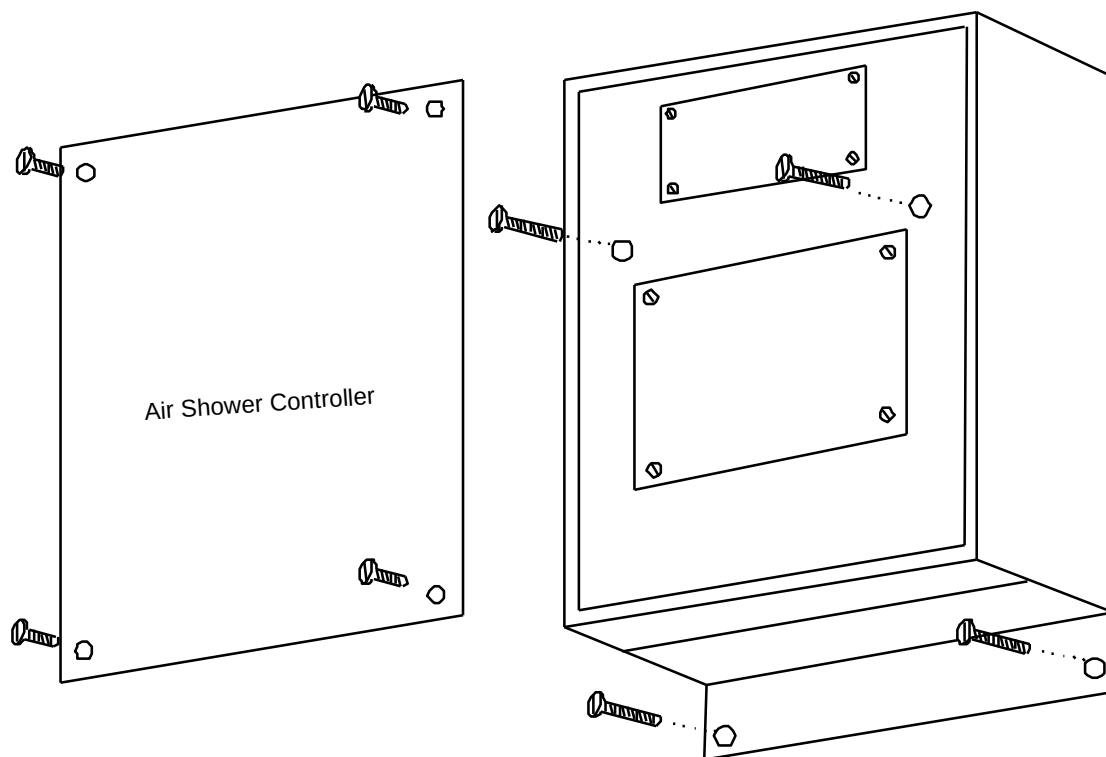


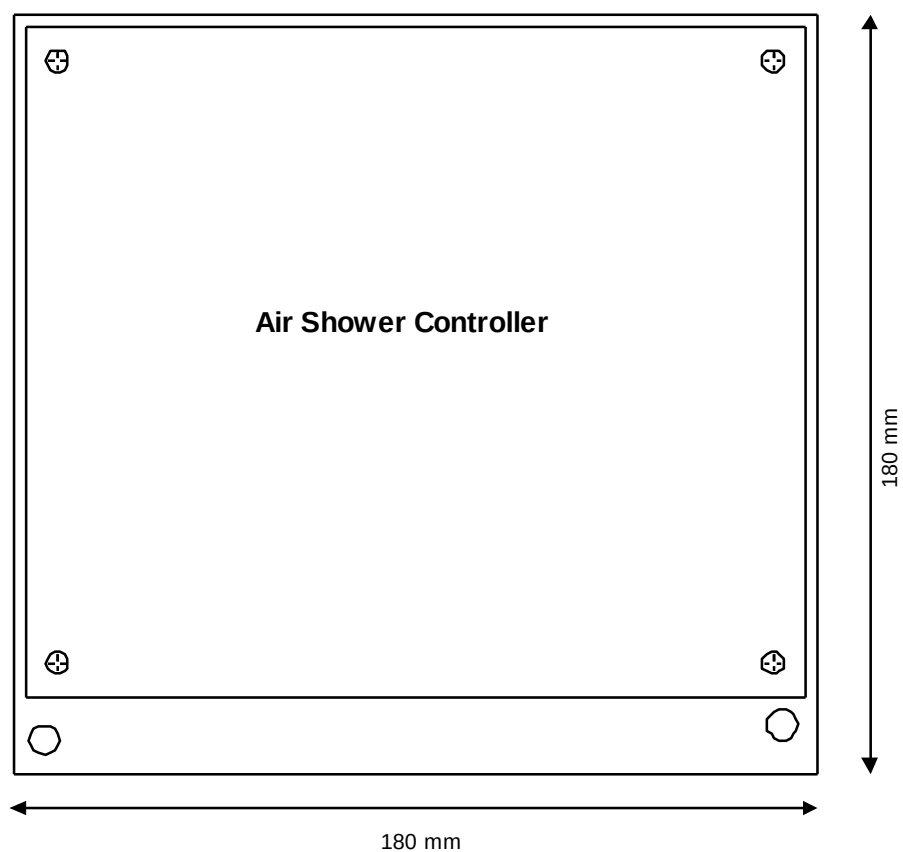
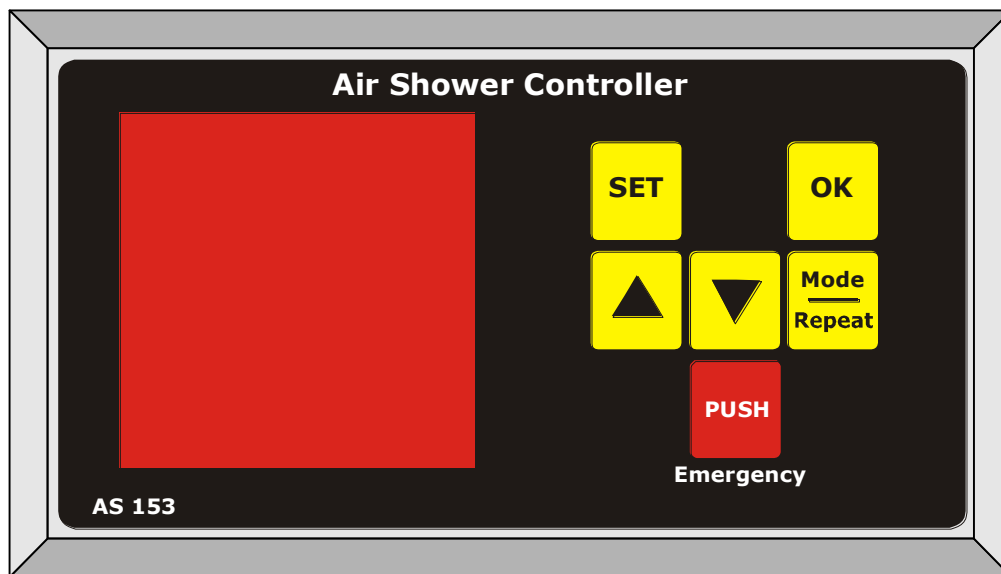
Fig (9.2) Front View Diagram of AS153C Controller:

Fig (9.3 Front View Diagram of Display Module:**10. Ordering Information:****I). Air shower controller:**

	X
<i>Model No</i>	<i>Future Use</i>
AS153	X

includes,

a. Controller:

AS153C	X	Model. No.
AS153C	X	AS153C-X

b. Display Module:

AS153D	X	Model. No.
AS153D	X	AS153D -X

Note:

For Further Information Contact:

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